①

問題2.5 (P25)

(a) $\int \psi^* \psi \, dx = A^2 \int |\psi_1|^2 + |\psi_2|^2 + \psi_1^* \psi_2 + \psi_2^* \psi_1 \, dx$

$= A^2 \int |\psi_1|^2 + |\psi_2|^2 \, dx = 2A^2 = 1, \quad A = \frac{1}{\sqrt{2}}$

~~(b) $C_1 \int \psi(x) \sin \frac{\pi}{a} x \, dx$~~

~~$= \int$~~

(b) $\Psi(x,t) = \frac{1}{\sqrt{2}} (\psi_1(x) e^{-iE_1 t/\hbar} + \psi_2(x) e^{-iE_2 t/\hbar})$

$= \frac{1}{\sqrt{a}} \left(\sin \frac{\pi x}{a} e^{-iE_1 t/\hbar} + \sin \frac{2\pi x}{a} e^{-iE_2 t/\hbar} \right)$

$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2} = n^2 \omega \hbar$

$= \frac{1}{\sqrt{a}} \left(\sin \frac{\pi x}{a} e^{-i\omega t} + \sin \frac{2\pi x}{a} e^{-i4\omega t} \right)$

(c) $|\Psi(x,t)|^2 = \psi^* \psi$

$= \frac{1}{a} \left[\sin^2 \frac{\pi x}{a} + \sin^2 \frac{2\pi x}{a} + \sin \frac{\pi x}{a} \sin \frac{2\pi x}{a} e^{i\omega t} e^{-i4\omega t} \right.$

$\left. + \sin \frac{2\pi x}{a} \sin \frac{\pi x}{a} e^{i4\omega t} e^{i\omega t} \right]$

$= \frac{1}{a} \left[\sin^2 \frac{\pi x}{a} + \sin^2 \frac{2\pi x}{a} + 2 \sin \frac{\pi x}{a} \sin \frac{2\pi x}{a} \cos 3\omega t \right]$

$\frac{d^2\psi}{dx^2} = -\frac{2mE}{\hbar^2}\psi$ 令 $k^2 = \frac{2mE}{\hbar^2}$

$= -k^2\psi$

$\psi(x) = A\sin kx + B\cos kx$

$\psi(0)$

①

习题 2.5 (P25)

(a) $\int \psi^* \psi$ 1

②

(c)

$\psi_1^* \psi_2$

$A =$

$$\langle x \rangle = \int_0^a \psi^* x \psi \, dx$$

$$= \frac{1}{a}\int_0^a x\left[\sin^2\frac{\pi x}{a} + \sin^2\frac{2\pi x}{a} + 2\sin\frac{\pi x}{a}\sin\frac{2\pi x}{a}\cos 3\omega t\right]dx$$

计算 $\int_0^a x\sin^2\frac{n\pi x}{a}\,dx = \int_0^a x\frac{1-\cos\frac{2n\pi x}{a}}{2}\,dx$

$= \frac{1}{2}\left[\int_0^a x\,dx - \int_0^a x\cos\frac{2n\pi x}{a}\,dx\right] = \frac{a^2}{4}$

$\langle x\rangle_n = \frac{a}{2}$

$-iE_2 t/\hbar$

$\hbar$)

$$\int_0^a x\sin\frac{\pi x}{a}\sin\frac{2\pi x}{a}\,dx = \frac{1}{2}\int_0^a x\left[\cos\frac{\pi x}{a} - \cos\frac{3\pi x}{a}\right]dx$$

$$= \left[\int_0^a x\cos\frac{\pi x}{a}\,dx - \int_0^a x\cos\frac{3\pi x}{a}\,dx\right]\frac{1}{2}$$

计算 $\int_0^a x\cos\frac{n\pi x}{a}\,dx = \int_0^a \frac{a}{n\pi}x\,d\sin\frac{n\pi x}{a}$

$$= \frac{a}{n\pi}\left[x\sin\frac{n\pi x}{a}\Big|_0^a - \int_0^a \sin\frac{n\pi x}{a}\,dx\right]$$

$$= \frac{a}{n\pi}\left[a\sin n\pi + \frac{a}{n\pi}\cos\frac{n\pi x}{a}\Big|_0^a\right] = \frac{a^2}{n^2\pi^2}(\cos n\pi - 1)$$

$$= \left[\frac{a^2}{\pi^2}(-2) - \frac{a^2}{9\pi^2}(-2)\right]\times\frac{1}{2} = \left[-\frac{2a^2}{9\pi^2} - \frac{2a^2}{\pi^2}\right]\frac{1}{2} = -\frac{16a^2}{9\pi^2}\times\frac{1}{2} = -\frac{8a^2}{9\pi^2}$$

$$\langle x\rangle = \frac{1}{2}\frac{a}{2} + \frac{1}{2}\frac{a}{2} + \frac{2}{a}\left(-\frac{8a^2}{9\pi^2}\right)\cos 3\omega t = \frac{a}{2} - \frac{16a}{9\pi^2}\cos(3\omega t)$$

平衡位置 $a/2$，角频率 $3\omega = 3\cdot\frac{\pi^2\hbar}{2ma^2}$ 振幅 $\frac{16a}{9\pi^2}$

(d) $\langle p\rangle = m\frac{d\langle x\rangle}{dt} = m\cdot\frac{3\pi^2\hbar}{2ma^2}\cdot\frac{16a}{9\pi^2}\sin 3\omega t$

$= \frac{8\hbar}{3a}\sin 3\omega t$

$\langle e\rangle$ $E_1 = \hbar\omega$

$E_2 = 4\hbar\omega$

$\langle H\rangle = \frac{5}{2}\hbar\omega$

① 习题 2.5 (P25)

(a) $\int \psi^* \psi \, dx$

② 习题 2.6

$A = \frac{1}{\sqrt{2}}$

$\psi(x,t) = \frac{1}{\sqrt{2}}\left(\sqrt{\frac{2}{a}} \sin\frac{\pi x}{a} e^{-i\omega t} + \sqrt{\frac{2}{a}} \sin\frac{2\pi x}{a} e^{-i\phi} e^{i\omega t + i\phi}\right)$

$|\psi(x,t)|^2 = \frac{1}{a}\left(\sin^2\frac{\pi x}{a} + \sin^2\frac{2\pi x}{a} + \sin\frac{\pi x}{a}\sin\frac{2\pi x}{a} e^{i\omega t - i\phi} + \sin\frac{\pi x}{a}\sin\frac{2\pi x}{a} e^{i\phi - i\omega t}\right)$

$= \frac{1}{a}\left(\sin^2\frac{\pi x}{a} + \sin^2\frac{2\pi x}{a} + 2\sin\frac{\pi x}{a}\sin\frac{2\pi x}{a}\cos(\omega t - \phi)\right)$

$\langle x \rangle = \int x |\psi(x,t)|^2 dx = \frac{a}{2} + \frac{2}{a}\int_0^a x \sin\frac{\pi x}{a}\sin\frac{2\pi x}{a}\cos(\omega t$

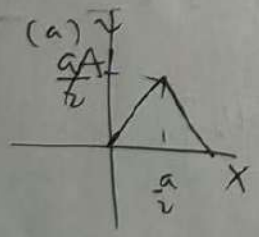
$= \frac{a}{2} - \frac{16a}{9\pi^2}\cos(\omega t - \phi)$

$\phi = \frac{\pi}{2}$ 时, $\langle x \rangle = \frac{a}{2} - \frac{16a}{9\pi^2}\sin\omega t$

$\phi = \pi$ 时 $\langle x \rangle = \frac{a}{2} + \frac{16a}{9\pi^2}\cos\omega t$

习题 2.7

(a)



$\int \psi^* \psi \, dx = 2\int_0^{\frac{a}{2}} (Ax)^2 dx = 2A^2 \int_0^{\frac{a}{2}} x^2 dx$

$= 2A^2 \frac{1}{3} x^3 \Big|_{\frac{a}{2}} = 2A^2 \cdot \frac{a^3}{8} \cdot \frac{1}{3} = \frac{A^2 a^3}{12} = 1$

$\Rightarrow A = \frac{2\sqrt{3}}{a^{\frac{3}{2}}}$

$\frac{12a}{a^3} \cdot \frac{\hbar^2}{m}$

(b) $\psi_n(x) = \sqrt{\frac{2}{a}} \sin\frac{n\pi x}{a}$ $\psi(x,0) = \sum C_n \psi_n(x)$

$E_n = \frac{n^2\pi^2\hbar^2}{2ma^2} = \hbar\omega$ $C_n = \int \psi(x,0)\psi_n^* dx$

$\psi_1^* \psi_2 + \psi_2^* \psi_1 \, dx$

$A = \frac{1}{\sqrt{2}}$

⑥

$C_n = \int \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a} \psi(x,0)\, dx$

$= \int_0^{\frac{a}{2}} \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a} A x \, dx + \int_{\frac{a}{2}}^{a} \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a} A(a-x)\, dx$

令 $t = a - x$　$t = x - \frac{a}{2}$　$a - x = a - (x + \frac{a}{2})$

$x = a - t$　$a - t + \frac{a}{2}$

$= A\sqrt{\frac{2}{a}} \int_0^{\frac{a}{2}} \sin \frac{n\pi x}{a} x \, dx + \int_0^{\frac{a}{2}} \sqrt{\frac{2}{a}} \sin \frac{n\pi (x + \frac{a}{2})}{a} (\frac{a}{2} - x)\, dx$

$\frac{n\pi}{a} + \frac{n\pi}{2}$

$\psi_n(a-x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi(a-x)}{a} = \sqrt{\frac{2}{a}} \sin(n\pi - \frac{n\pi x}{a})$

$= (-1)\sqrt{\frac{2}{a}} \sin(\frac{n\pi x}{a} - n\pi)$

$= (-1)^{n+1} \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a} = (-1)^{n+1} \psi_n(x)$

$\psi(x,0)$ 为偶函数 $(\frac{a}{2})$ 故 $C_n = 0$ (n 为偶数时)

故宏观 n 为奇数

$$\frac{\int_{\frac{a}{2}}^{0} \sqrt{\frac{2}{a}} \sin \frac{n\pi(a-t)}{a} A t \, d(a-t)}{(n\pi - \frac{n\pi x}{a})}$$

$$\int_0^{\frac{a}{2}} \sqrt{\frac{2}{a}} \sin \frac{n\pi(a-x)}{a} A x \, dx$$

$\sin n\pi x / a$

故 $C_n = \begin{cases} \frac{4\sqrt{6}}{n^2\pi^2} \sin \frac{n\pi}{2}, & n \text{ 为奇} \\ 0, & n \text{ 为偶} \end{cases}$

$C_n = 2A\sqrt{\frac{2}{a}} \int_0^{\frac{a}{2}} \sin \frac{n\pi x}{a} x \, dx$

$= 2A\sqrt{\frac{2}{a}} \left[-x \frac{\cos}{n\pi} \cos \frac{n\pi x}{a} \Big|_0^{\frac{a}{2}} + \int_0^{\frac{a}{2}} \cos \frac{n\pi x}{a} dx \right]$

$= 2A\sqrt{\frac{2}{a}} \left(-\frac{a}{2} \frac{a}{n\pi} \cos \frac{n\pi}{2} + \frac{a^2}{n^2\pi^2} \sin \frac{n\pi}{2} \right)$　n 为奇数

$= 2A\sqrt{\frac{2}{a}} \cdot \frac{a^2}{n^2\pi^2} \sin \frac{n\pi}{2} = \frac{4\sqrt{3}}{a\sqrt{a}} \frac{\sqrt{2}}{\sqrt{a}} \frac{a^2}{n^2\pi^2} \sin \frac{n\pi}{2} = \frac{4\sqrt{6}}{n^2\pi^2} \sin \frac{n\pi}{2}$

(b)

$$\Psi(x,t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-iE_n t/\hbar}$$

$$= \sum_{n=1,3,5,\ldots} \frac{8\sqrt{3}}{\pi^2\sqrt{a}} \frac{1}{n^2} \sin\frac{n\pi}{2} \sin\frac{n\pi x}{a} e^{-i\frac{n^2\pi^2\hbar^2}{2ma^2}t}$$

(c) $P_1 = C_1^2 = \left(\frac{4\sqrt{6}}{\pi^2}\right)^2 = \frac{96}{\pi^4}$

(d) $\langle E \rangle = \sum_{n=1,3,5,\ldots} C_n^2 E_n = \frac{96}{n^4\pi^4} \sin^2\frac{n\pi}{2} \frac{n^2\pi^2\hbar^2}{2ma^2}$

$$= \Sigma \frac{96\hbar^2}{n^2\pi^2} \frac{1}{2ma^2} = \frac{96\hbar^2}{2ma^2\pi^2} \sum_{n=1,3,5,\ldots} \frac{1}{n^2}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} = \left(1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots\right)$$

$$= \left(\frac{1}{1^2} + \frac{1}{3^2} + \cdots\right) + \left(\frac{1}{2^2} + \frac{1}{4^2} + \cdots\right)$$

$$= \zeta_{奇} + \frac{1}{4}\zeta = \frac{\pi^2}{6} = \zeta \quad \zeta_{奇} = \frac{\pi^2}{8}$$

$$\text{故} \langle E \rangle = \frac{6\hbar^2}{ma^2}$$

$$\langle H \rangle = \int \psi^* \langle H \rangle \psi \, dx = \int \psi^* \left(-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2}\right) dx$$

$$= \int_0^a \psi^* \frac{\hbar^2}{2m} 2A\,\delta\left(x - \frac{a}{2}\right) dx = 2 \cdot \frac{A\hbar^2}{m}\psi\left(\frac{a}{2}\right) = \frac{aA^2\hbar^2}{2m}$$

$$= \frac{6\hbar^2}{ma^2}$$

⑥

2.10 P33

(1) $\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar}x^2}$

$a_+ \psi_0(x) = \psi_1(x)$

$a_+ \psi_1(x) = \sqrt{2}\psi_2(x)$

$\xi = \sqrt{\frac{m\omega}{\hbar}}x \quad \psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{\xi^2}{2}}$

$a_+ = \frac{1}{\sqrt{2\hbar m\omega}}(-ip + m\omega x) = \left(-\hbar\frac{d}{dx} + m\omega x\right)\frac{1}{\sqrt{2\hbar m\omega}}$

$= \frac{1}{\sqrt{2}}\left(\xi - \sqrt{\frac{\hbar}{m\omega}}\frac{d}{dx}\right)$

$\psi_1(\xi) = \frac{1}{\sqrt{2}}\left(\xi - \frac{d}{d\xi}\right)\psi_0$

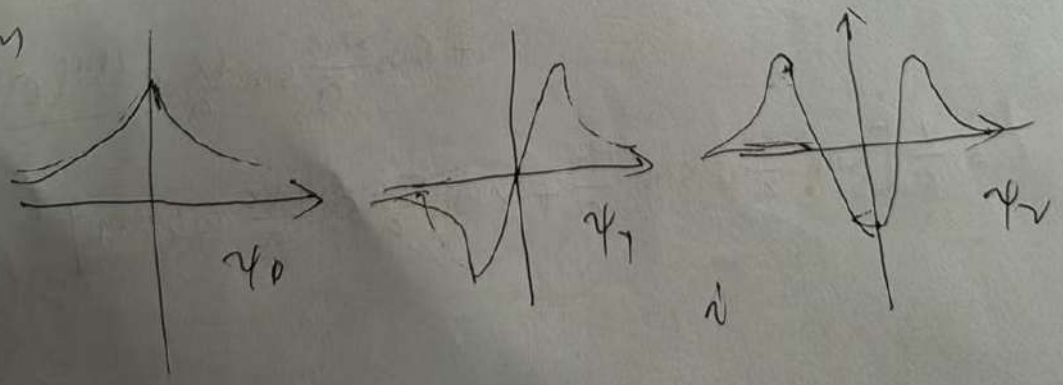
$= \frac{1}{\sqrt{2}}\xi\psi_0 + \frac{1}{\sqrt{2}}\xi\psi_0 = \sqrt{2}\xi\psi_0$

$\psi_2 = \frac{1}{\sqrt{2}}a_+\psi_1 = \frac{1}{\sqrt{2}}\left(\xi - \frac{d}{d\xi}\right)\sqrt{2}\xi\psi_0$

$= \frac{1}{\sqrt{2}}\left[\sqrt{2}\xi^2\psi_0 - \psi_0 - \sqrt{2}\xi^2\psi_0\right] = \frac{1}{\sqrt{2}}(-1)\psi_0$

$\psi_2(x) = \frac{1}{\sqrt{2}}\left(\frac{2m\omega x^2}{\hbar} - 1\right)\left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega x^2}{2\hbar}}$

(2)



$\frac{d\langle p\rangle}{dt} = \langle -\frac{\partial V}{\partial x}\rangle$

⑩ (1) $\int \psi_0^* \psi_1 dx = 0 \quad \int \psi_2^* \psi_3 dx = 0$

$$\int \psi_0^* \psi_2 dx = \int \frac{1}{\sqrt{2}}\left(\frac{2m\omega x^2}{\hbar} - 1\right)\left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{2}} e^{-\frac{m\omega x^2}{\hbar}} dx$$

$$= \frac{1}{\sqrt{2}}\left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{2}}\left[\int \frac{2m\omega}{\hbar} x^2 e^{-\frac{m\omega x^2}{\hbar}} dx - \int e^{-\frac{m\omega x^2}{\hbar}} dx\right]$$

$$= \frac{1}{\sqrt{2}}\left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{2}}\left[\frac{2m\omega}{\hbar}\sqrt{\frac{\pi\hbar}{m\omega}}\frac{\hbar}{2m\omega} - \sqrt{\frac{\pi\hbar}{m\omega}}\right] = 0$$

— 2.12 P33

$$\psi_n = \frac{1}{\sqrt{n!}}(a_+)^n \psi_0$$

$$a_+ = (-ip + m\omega x)\frac{1}{\sqrt{2\pi\hbar m\omega}} \quad p = i\sqrt{\frac{m\hbar\omega}{2}}(a_+^\dagger - a_-)$$

$$a_- = (ip + m\omega x)\frac{1}{\sqrt{2\pi\hbar m\omega}} \quad x = \sqrt{\frac{\hbar}{2m\omega}}(a^+ + a^-)$$

$$x^2 = \frac{\hbar}{2m\omega}\left((a^+)^2 + (a^-)^2 + (a_+a_-) + (a_-a_+)\right)$$

$$\langle x \rangle = \int \psi^* \sqrt{\frac{\hbar}{2m\omega}}\, a_+^\dagger \psi_n dx + \int \psi^* \sqrt{\frac{\hbar}{2m\omega}}\, a_- \psi_n dx$$

$$= 0$$

$a_+\psi_n = \sqrt{n+1}\,\psi_{n+1}$

$a_-\psi_{n+1} = \sqrt{n+1}\,\psi_n$

~~$\langle x^2 \rangle \quad \int \frac{\hbar}{2m\omega}\left[\int \psi_n^* (a^+)^2 \psi_n dx + \int \psi_n^* (a_-)^2 \psi_n dx\right]$~~

$$\langle x^2 \rangle = \frac{\hbar}{2m\omega}\int \psi_n^* (a_-a_+ + a_+a_-)\psi_n dx$$

$$= \frac{\hbar}{2m\omega}(n+1+n) = \left(\frac{1}{2}+n\right)\frac{\hbar}{m\omega} \quad \frac{1}{2}m\omega^2$$

$\langle p \rangle = 0 \quad \langle p^2 \rangle = \frac{1}{2}m\omega\hbar(n+\frac{1}{2})$

$\langle T \rangle = \frac{\langle p^2 \rangle}{2m} = \frac{\hbar\omega}{2}(n+\frac{1}{2})$

$\Delta x_n = \sqrt{\frac{\hbar}{m\omega}(n+\frac{1}{2})} \quad \Delta p = \sqrt{m\hbar\omega(n+\frac{1}{2})}$

$\Delta x \cdot p = \hbar(n+\frac{1}{2}) \geq \frac{\hbar}{2}$

$$= \frac{2\psi}{2\hbar}\sqrt{\frac{\hbar}{2m\omega}}$$

$$\frac{d\langle p \rangle}{dt} = \left\langle -\frac{\partial V}{\partial x} \right\rangle$$

(8) 2.13

(a) $\int \psi^* \psi \, dx = |A^2| \int 9|\psi_0(x)|^2 dx + \int 16|\psi_1(x)|^2 dx$

$= A^2(25) = 1 \quad A = \frac{1}{5}$

(b) $\psi(x,t) = \frac{1}{5}(3\psi_0 e^{-i\frac{\hbar\omega}{2}t/\hbar} + 4\psi_1 e^{-i\frac{3\hbar\omega}{2}t/\hbar})$

$= \frac{1}{5}(3\psi_0 e^{-i\omega t/2} + 4\psi_1 e^{-i3\omega t/2})$

$|\psi(x,t)|^2 = \psi^*\psi = \frac{1}{25}(9\psi_0^2 + 16\psi_1^2 + 24\psi_0\psi_1 \cos 3\omega t)$

(c) $x = \sqrt{\frac{\hbar}{2m\omega}}(a_- + a_+)$

$p = i\sqrt{\frac{m\omega\hbar}{2}}(a_- - a_+)$

$a_+\psi_0 = \sqrt{1}\,\psi_1$

$a_-\psi_0 = 0$

$\langle x \rangle = \sqrt{\frac{\hbar}{2m\omega}} \int \psi^* (a_+ + a_-) \psi \, dx$

$= \sqrt{\frac{\hbar}{2m\omega}} \int \left[\frac{3}{5}\psi_0^* e^{+i\omega t/2} + \frac{4}{5}\psi_1^* e^{+i3\omega t/2}\right](a_- + a_+)$

$\left[\frac{3}{5}\psi_0 e^{-i\omega t/2} + \frac{4}{5}\psi_1 e^{-i3\omega t/2}\right]$

$= \sqrt{\frac{\hbar}{2m\omega}} \int \frac{12}{25}\psi_1^* \psi e^{-i\omega t} + \frac{12}{25}\psi_0^* \psi e^{-i\omega t} \, dx$

$= \frac{24}{25}\sqrt{\frac{\hbar}{2m\omega}}\cos\omega t \qquad \langle p \rangle = m\frac{d\langle x\rangle}{dt} = -\frac{24}{25}\sqrt{\frac{m\hbar\omega}{2}}\sin\omega t$

$\frac{d\langle p\rangle}{dt} = \left\langle -\frac{\partial V}{\partial x} \right\rangle \qquad \frac{1}{2}m\omega^2 = \left[-\frac{\hbar^2}{2m}\frac{d^2}{dx} + V(x)\right]$

$\frac{1}{2}kx^2$

⑨ 2.14

$\psi_0 = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar}x^2}$ $\quad \omega' = 2\omega$

$E_n = \hbar\omega'\left(n+\frac{1}{2}\right) = \hbar\omega(2n+1)$ $\quad P\left(\frac{\hbar\omega}{2}\right) = 0$

$\psi_0'(x) = \left(\frac{m\omega'}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega' x^2}{2\hbar}}$

~~$P(E = \hbar\omega') = (\sum_{n=0}^{+\infty} \psi_0$~~

$\psi_0 = C_0\psi_0' + C_1\psi_1' + \cdots + C_n\psi_n'$

$C_0 = \left|\int \psi_0'^{*}\psi_0\, dx\right|^2 = \left|\left(\frac{\pi\omega'}{\pi\hbar}\right)^{\frac{1}{4}} \left(\frac{\pi\omega}{\pi\hbar}\right)^{\frac{1}{4}} \int e^{-\frac{m(\omega+\omega')}{2\hbar}x^2} dx\right|^2$

$= \left[\sqrt{\frac{2\pi\hbar}{m(\omega+\omega')}} \frac{\pi^{\frac{1}{4}}\omega'^{\frac{1}{4}}\omega^{\frac{1}{4}}\sqrt{\pi}}{(\pi\hbar)^{\frac{1}{2}}} \sqrt{\frac{m}{\pi\hbar}} (\omega\omega')^{\frac{1}{4}}\right]^2$

$= \frac{2\sqrt{\omega\omega'}}{\omega+\omega'} = \frac{2\sqrt{2}}{3} \approx 0.943$